Algebra Review: Unit 8 — Quadratics

Station 1 of 4: Solving by Factoring and Square Roots

Name: _____ Date: _____

Solve each equation. Give exact answers; leave radicals in simplest form.

1. Solve: $x^2 + 5x + 6 = 0$

2. Solve: $x^2 - 7x + 12 = 0$

3. Solve: $x^2 - x - 20 = 0$

4. Solve: $2x^2 - 8x = 0$

5. Solve: $x^2 - 16 = 0$

6. Solve: $x^2 = 20$

7. Solve: $3x^2 - 12 = 0$

8. Solve: $(x - 3)^2 = 25$

9. Solve: $2x^2 + x - 6 = 0$

10. Solve: $(x + 1)^2 = 18$

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Station 2 of 4: Completing the Square

Name: _____ Date: _____

Complete the square. Give exact answers.

1. What number completes the square? $x^2 + 8x + _____$

2. What number completes the square? $x^2 - 10x + _____$

3. What number completes the square? $x^2 + 3x + _____$

4. Solve by completing the square: $x^2 + 6x = 7$

5. Solve by completing the square: $x^2 - 8x + 7 = 0$

6. Solve by completing the square: $x^2 + 4x - 1 = 0$

7. Solve by completing the square: $x^2 - 2x - 5 = 0$

8. Solve by completing the square: $x^2 + x - 6 = 0$

9. Solve by completing the square: $2x^2 + 12x - 14 = 0$

10. Write $y = x^2 - 6x + 5$ in vertex form.

Algebra Review: Unit 8 — Quadratics

Station 3 of 4: The Quadratic Formula and the Discriminant

Name: _____ Date: _____

Use $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. *Give exact answers in simplest form.*

1. Find the discriminant of $2x^2 - 3x - 5 = 0$ and tell how many real solutions it has.

2. Find the discriminant of $x^2 + 4x + 4 = 0$ and tell how many real solutions it has.

3. Find the discriminant of $x^2 + 2x + 5 = 0$ and tell how many real solutions it has.

4. Solve: $x^2 - 3x - 10 = 0$

5. Solve: $2x^2 + 5x - 3 = 0$

6. Solve: $x^2 + 6x + 4 = 0$

7. Solve: $x^2 - 4x + 1 = 0$

8. Solve: $3x^2 - 2x - 2 = 0$

9. Solve: $4x^2 - 4x + 1 = 0$

10. Solve: $2x^2 - 6x + 3 = 0$

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Station 4 of 4: Graphs of Quadratic Functions and Applications

Name: _____ Date: _____

Find the vertex, axis of symmetry, and other features. Include units in word problems. Diagrams are not drawn to scale.

1. For $y = x^2 - 4x + 3$, find the vertex, axis of symmetry, direction of opening, and y-intercept.

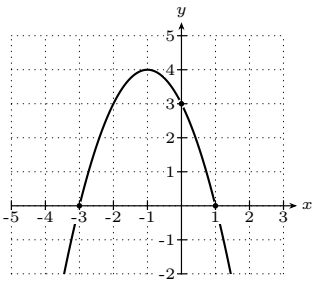
2. For $y = -x^2 + 6x - 5$, find the vertex, axis of symmetry, direction of opening, and y-intercept.

3. For $y = 2x^2 + 8x + 5$, find the vertex, axis of symmetry, direction of opening, and y-intercept.

4. Find the vertex and x-intercepts of $y = (x - 1)^2 - 4$.

5. Find the vertex and x-intercepts of $y = -2(x + 3)^2 + 8$. Does it open up or down?

6. The graph of a parabola is shown. Find (a) the vertex, (b) the axis of symmetry, (c) the x-intercepts, and (d) the y-intercept.



7. Find the vertex of $y = -3x^2 + 12x - 7$.

8. A ball is thrown so its height in feet is $h = -16t^2 + 32t + 48$, where t is in seconds. Find the maximum height and when it occurs.

9. For the ball in the previous problem, how long until it hits the ground?

10. A store's revenue (in dollars) is $R = -5p^2 + 100p$, where p is the price of an item. What price gives the maximum revenue, and what is that revenue?

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Answer Key

Station 1: Solving by Factoring and Square Roots

1. $x = -3, x = -2$
2. $x = 3, x = 4$
3. $x = -4, x = 5$
4. $x = 0, x = 4$
5. $x = -4, x = 4$
6. $x = \pm 2\sqrt{5}$
7. $x = -2, x = 2$
8. $x = -2, x = 8$
9. $x = -2, x = \dfrac{3}{2}$
10. $x = -1 \pm 3\sqrt{2}$

Station 2: Completing the Square

1. 16
2. 25
3. $\dfrac{9}{4}$
4. $x = -7,\ x = 1 \quad (x+3)^2 = 16$
5. $x = 1,\ x = 7 \quad (x-4)^2 = 9$
6. $x = -2 \pm \sqrt{5} \quad (x+2)^2 = 5$
7. $x = 1 \pm \sqrt{6} \quad (x-1)^2 = 6$
8. $x = -3,\ x = 2 \quad \left(x + \frac{1}{2}\right)^2 = \dfrac{25}{4}$
9. $x = -7,\ x = 1 \quad$ *divide by 2 first*
10. $y = (x-3)^2 - 4$

Station 3: The Quadratic Formula and the Discriminant

1. $b^2 - 4ac = 49$; two real solutions
2. $b^2 - 4ac = 0$; one real solution
3. $b^2 - 4ac = -16$; no real solutions
4. $x = -2, x = 5$
5. $x = -3, x = \dfrac{1}{2}$
6. $x = -3 \pm \sqrt{5}$
7. $x = 2 \pm \sqrt{3}$
8. $x = \dfrac{1 \pm \sqrt{7}}{3}$
9. $x = \dfrac{1}{2}$
10. $x = \dfrac{3 \pm \sqrt{3}}{2}$

Station 4: Graphs of Quadratic Functions and Applications

1. $(2, -1)$; $x = 2$; opens up; y-int 3
2. $(3, 4)$; $x = 3$; opens down; y-int -5
3. $(-2, -3)$; $x = -2$; opens up; y-int 5
4. $(1, -4)$; $x = -1,\ x = 3 \quad (x-1)^2 = 4$
5. $(-3, 8)$; $x = -5,\ x = -1$; opens down $\quad (x+3)^2 = 4$
6. (a) $(-1, 4)$ (b) $x = -1$ (c) $x = -3,\ x = 1$ (d) $(0, 3)$
7. $(2, 5)$
8. Maximum height 64 ft at $t = 1$ s
9. $t = 3$ seconds $\quad -16(t-3)(t+1) = 0$
10. Price \$10; revenue \$500